

# Routing-Level Signatures of Cognitive Efficiency: Only Inter-Subject Routing Consistency Survives FDR, with a Pairwise-Correlation Caveat

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**Abstract**—We jointly test four pre-specified routing-level candidate signatures of cognitive efficiency (activation-distribution sparsity, canonical task-network engagement, inter-subject routing consistency, and modularity) in 100 Human Connectome Project Young Adult participants across all seven task-fMRI paradigms, with efficiency groups defined by working-memory behavior and Benjamini–Hochberg (BH) FDR control. Only inter-subject routing consistency survives FDR: under a subject-level leave-one-out (LOO) statistic, high-efficiency participants route more like one another ( $d = +0.52$ , 95% CI  $[+0.12, +0.92]$ ,  $p = 0.011$ ,  $p_{\text{FDR}} = 0.033$ ; label permutation  $p = 0.018$ ;  $\text{BF}_{10} = 4.0$ ), and consistency correlates with the continuous efficiency score ( $r = 0.36$ ,  $p = 0.0002$ ). The effect, however, is carried by the Working Memory task that also defines the efficiency score; without it,  $d = +0.19$  ( $p = 0.34$ ). Sparsity ( $d = -0.26$ ), task-network engagement (2/7 canonical matches, at chance) and modularity ( $d = +0.19$ ) are not supported. On the same consistency data, the conventional within-group pairwise-correlation  $t$ -test reports  $p = 1.0 \times 10^{-30}$ , overstating significance by about 28 orders of magnitude because it treats 2450 correlated pairs as independent; we show the inflation directly and give a drop-in subject-level replacement. A cross-domain comparison with three Mixture-of-Experts (MoE) language models finds cortex about  $3.3\times$  more concentrated than MoE expert use pooled over layers, but single MoE layers can be far more concentrated, so that comparison depends on the level of pooling. Routing consistency is at most a candidate marker of efficient cognition; testing it requires an efficiency measure that is independent of the imaged tasks.

**Index Terms**—neural efficiency, multi-task fMRI, inter-subject routing consistency, pairwise-correlation inference, FDR

## I. INTRODUCTION

Cognitive efficiency varies substantially between individuals and declines with age [1], [2], with pronounced deficits in Alzheimer’s disease [3]–[5] and schizophrenia [6]. Translating this heterogeneity into individual-subject prediction is a central goal of “precision neuroimaging” [7], [8], and task fMRI is an attractive substrate because it probes how information is routed across functional networks, the system whose selective, metabolically bounded engagement is thought to underlie efficient cognition [2], [9].

The neural efficiency hypothesis proposes that more capable brains accomplish tasks with fewer neural resources [2], [10], and sparse coding theory formalizes the idea that information is carried by selective activation of small neuronal subsets [11], [12]. A routing-based account must also say *which* circuits are engaged for which task, *how consistent* that routing is across individuals, and *how modular* the resulting organization is, each with its own clinical reading, for example loss of network segregation in Alzheimer’s disease [3], [4] or diffuse recruitment during working memory in schizophrenia [6]. Individual-differences studies have usually examined one of these metrics at a time: modularity as a marker of plasticity [13], [14], connectivity fingerprinting [7], [8], or activation extent [10].

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To our knowledge the four properties (sparser, more task-selective, more consistent, more modular) have not been tested jointly in one sample with a behavior-derived efficiency label and multiple-comparison control.

We also compare cortex with three Mixture-of-Experts (MoE) language models [15], [16], which obtain computational efficiency by sparse expert routing [17]. Earlier brain-network comparisons concern representational similarity [18]–[20] rather than routing.

The paper makes three contributions. (1) In 100 HCP participants, with the HCP-MMP1.0 parcellation, task-canonical GLM contrasts, Louvain community detection and BH-FDR-controlled joint testing, one of the four candidates, inter-subject routing consistency, survives FDR, and permutation, rank, continuous-score and Bayesian analyses agree, although the effect is carried by the Working Memory task on which efficiency is defined; sparsity, canonical task-network engagement and modularity are not supported. (2) We quantify an inference pitfall that the consistency analysis invites: the within-group pairwise-correlation  $t$ -test treats  $n(n-1)/2$  non-independent pairs as independent and, on our data, overstates significance by about 28 orders of magnitude relative to subject-level and permutation inference (Fig. 2); we give the subject-level statistic as a drop-in replacement. (3) Cortex is more concentrated than MoE expert use pooled over layers, but single MoE layers can be more concentrated than cortex, so cross-domain efficiency comparisons must state the level of pooling.

## II. METHODS

### A. Participants, Data, and Preprocessing

We used HCP Young Adult task fMRI [21] from 100 healthy adults (53 female; open-access age bands 22–25: 17, 26–30: 42, 31–35: 40, > 35: 1). The sample is the first 100 participants, in ascending subject ID, of a list of HCP-YA participants with complete data for all seven tasks in both phase-encoding directions (LR and RL); no further selection was applied, and no motion threshold was used for selection. Head motion was nonetheless low: mean relative RMS displacement was 0.089 mm (SD 0.030), no run exceeded 0.5 mm (maximum 0.36 mm), and motion did not differ between efficiency groups ( $p = 0.64$ ). We used the MSMAll-aligned, minimally preprocessed CIFTI time series (`tfMRI*_Atlas_MSMAll.dtseries.nii`) [22], [23], without ICA-FIX denoising. HCP-YA contains twins and siblings; family identifiers are restricted data that we did not use, so family structure is not modelled (Sec. IV-D).

### B. Parcellation and Activation Extraction

Each run’s grayordinate time series was averaged within the 360 cortical parcels of HCP-MMP1.0 [24]. For each run we fitted a GLM to the parcel time series containing only the HCP block regressors of that task, convolved with the SPM canonical double-gamma haemodynamic response ( $TR = 0.72$  s), a discrete-cosine high-pass basis (128 s) and an intercept. Contrasts followed the HCP task definitions: 2-back

vs. 0-back (WM); all five movements vs. fixation (MOTOR); story vs. math (LANGUAGE); mental vs. random motion (SOCIAL); relational vs. match (RELATIONAL); fearful faces vs. shapes (EMOTION); win vs. loss (GAMBLING). The LR and RL contrast vectors were averaged and  $z$ -scored across parcels, giving a  $100 \times 7 \times 360$  activation array; all 700 subject  $\times$  task cells were extracted, and design condition numbers were at most 1.7. Each parcel was assigned to one of the 12 Cole-Anticevic networks [25] by modal vote, and the 12 were collapsed to seven Yeo-like networks [26]: {Visual1, Visual2}  $\rightarrow$  VIS; {Somatomotor, Auditory}  $\rightarrow$  SMN; {Dorsal-Attention}  $\rightarrow$  DAN; {Cingulo-Opercular, Language, Ventral-Multimodal}  $\rightarrow$  VAN; {Orbito-Affective}  $\rightarrow$  LIM; {Frontoparietal, Posterior-Multimodal}  $\rightarrow$  FPN; {Default}  $\rightarrow$  DMN (60, 54, 23, 83, 6, 57 and 77 parcels).

### C. Efficiency Score

Participants were split at the median of a working-memory accuracy-to-reaction-time ratio,  $E_i = \text{Acc}_i / (\text{RT}_i / \overline{\text{RT}})$ , with  $\text{Acc}_i$  the mean 0-back/2-back accuracy and  $\text{RT}_i$  the mean median response time, into high- and low-efficiency groups ( $n = 50$  each). The score is independent of the neuroimaging measures, being computed from behavior alone, but not of the task battery: Working Memory is also one of the seven fMRI paradigms. Its run-to-run reliability (LR vs. RL) was  $r = 0.80$  (Spearman–Brown 0.89), and 84% of participants fall on the same side of the median in both runs. Because a median split discards information, every primary measure is also related to the continuous  $E_i$  (Sec. III-E).

### D. Hypotheses, Primary Outcomes, and Inference

Each hypothesis has one primary outcome and one test, computed with one value per participant. **H1** (efficient brains are sparser): composite sparsity averaged over the seven tasks, Student  $t$ . **H2** (task-selective engagement): the descriptive canonical-match count below and a group test of each participant’s mean entropy selectivity (Welch  $t$ ). **H3** (more consistent routing): subject-level LOO correlation, Welch  $t$ . **H4** (more modular): Newman’s  $Q$ , Student  $t$ . The pre-specified primary BH-FDR family is {H1, H3, H4}; because the pre-specified plan had no group test for H2, it enters only a four-test sensitivity family. Secondary families (the four H1 metrics, the three H4 metrics, the seven per-task H1 tests and the six continuous-score correlations) are corrected within themselves. For each primary test we report Cohen’s  $d$  with a 95% CI, a two one-sided test (TOST) of equivalence to  $|d| < 0.5$  [27], and a JZS Bayes factor  $\text{BF}_{10}$  (Cauchy prior scale  $\sqrt{2}/2$ ) [28]. Testing H1 on all 700 subject  $\times$  task values as if independent would violate the independence requirement that motivates Sec. II-F; the subject-level test respects it.

### E. Activation-Distribution Sparsity

Three indices describe how concentrated a task map is across parcels: the Gini coefficient of  $u_i = |a_i|$ ,  $G = \sum_{i,j} |u_i - u_j| / (2n \sum_i u_i)$ ; an entropy index  $S = 1 - H / \ln B$

from a  $B = 50$ -bin histogram of the  $z$ -scored values; and the share of  $\sum_i u_i$  carried by the top 10% of parcels. The composite is their mean. Because maps are  $z$ -scored within subject and task, these indices measure the spatial concentration of the activation distribution, not activation extent, amplitude or metabolic cost; we call the construct activation-distribution sparsity and do not equate it with neural resource consumption.  $z$ -scoring also makes  $L_0$  sparsity and supra-threshold region counts constant, so they are not used.

#### F. Routing Consistency and the Subject-Level Statistic

Each participant’s routing profile  $\mathbf{x}_i$  is the flattened  $7 \times 7$  task-by-network matrix of network-mean activations. The primary H3 statistic is

$$r_i^{\text{LOO}} = \text{corr}\left(\mathbf{x}_i, \frac{1}{|g_i|-1} \sum_{j \in g_i, j \neq i} \mathbf{x}_j\right), \quad (1)$$

where  $g_i$  is  $i$ ’s group; the 100 subject-level LOO statistics enter a Welch  $t$ -test. In pseudocode: for  $i$  in group:  $r[i] = \text{corr}(\mathbf{x}[i], \text{mean}(\mathbf{x}[\text{group} \setminus i]))$ ;  $\text{welch\_t}(r[\text{high}], r[\text{low}])$ . The LOO templates of members of the same group overlap, so the  $r_i$  are not strictly independent; a label-permutation test (10 000 permutations, statistic = difference of within-group mean pairwise  $r$ ), which makes no independence assumption, is the confirmatory check. For comparison we also run the analysis a practitioner would reach for: all 1225 within-group pairwise correlations per group pooled into an independent-samples  $t$ -test.

#### G. Task Selectivity and Modularity

Task selectivity is  $S_t = 1 - H_t / \log_2 K$ , with  $H_t$  the entropy of the absolute network-mean activations normalized to proportions over  $K$  networks ( $K = 7$ , or  $K = 12$  at native resolution). Canonical expectations were fixed before analysis at Yeo-7 (WM, RELATIONAL: FPN or DAN; MOTOR: SMN; LANGUAGE: FPN or VAN; SOCIAL: DMN; EMOTION: LIM; GAMBLING: FPN); at 12 networks each expectation is carried over to the Cole-Anticevic networks that collapse into it, so no new choice is made. A task matches if its highest signed network mean is an expected network; chance is estimated by permuting network labels.

For H4, each participant’s  $360 \times 360$  matrix  $A$  holds Pearson correlations between parcels’ seven-task activation vectors. This is task-activation-profile similarity with seven observations per edge, not time-series functional connectivity. Louvain [29] (resolution 1, seed 0) is run on  $|A|$ : two parcels whose task profiles are strongly anti-correlated are coupled by the same task factor with opposite sign, so we treat them as related rather than unrelated; a positive-edge-only graph is a sensitivity check. From the partition  $c$  we compute Newman’s  $Q = \frac{1}{2m} \sum_{ij} [|A_{ij}| - k_i k_j / 2m] \delta(c_i, c_j)$ , segregation  $(\bar{w}_{\text{within}} - \bar{w}_{\text{between}}) / (\bar{w}_{\text{within}} + \bar{w}_{\text{between}})$  with  $\bar{w}$  the mean  $|A_{ij}|$  within and between communities, and the mean participation coefficient  $P_i = 1 - \sum_c (k_{ic} / k_i)^2$  [30]. Partition stability is assessed over 20 Louvain seeds.

TABLE I: Correspondence between fMRI tasks and MoE text categories. The pairing is an analogy, not a matched manipulation; RISK–GAMBLING is the weakest.

| fMRI task      | Text category | Source (selection)              |
|----------------|---------------|---------------------------------|
| Working Memory | Mathematical  | GSM8K word problems             |
| Motor          | Procedural    | RACE passages (instructional)   |
| Language       | Narrative     | RACE passages (story)           |
| Social         | Social        | SNLI premises (social keywords) |
| Relational     | Scientific    | RACE passages (expository)      |
| Emotion        | Emotional     | SST-2 sentences                 |
| Gambling       | Risk decision | SNLI premises (risk keywords)   |

TABLE II: Activation-distribution sparsity by efficiency group, one value per participant (mean over seven tasks). Negative  $d$ : the high-efficiency group is less sparse.  $p_{\text{FDR}}$  is within the four-metric H1 family.

| Metric              | High  | Low   | $d$   | $p$   | $p_{\text{FDR}}$ |
|---------------------|-------|-------|-------|-------|------------------|
| Gini coefficient    | 0.445 | 0.445 | −0.05 | 0.80  | 0.80             |
| Entropy sparsity    | 0.157 | 0.163 | −0.35 | 0.082 | 0.33             |
| Concentration ratio | 0.288 | 0.290 | −0.21 | 0.30  | 0.40             |
| Composite           | 0.297 | 0.299 | −0.26 | 0.19  | 0.39             |

#### H. Mixture-of-Experts Models

We analyzed Switch-base-8 (8 experts, Top-1) [16], Qwen1.5-MoE-A2.7B (60 routed experts, Top-4) [31] and DeepSeek-MoE-16B-base (64 routed experts, Top-6) [32]. Text came from RACE [33], SNLI [34], SST-2 [35] and GSM8K [36], sorted into seven categories each paired with one fMRI task (Table I); 618 samples were collected and up to 50 per category (318 sequences,  $\leq 128$  tokens) were run through each model. Routing was read from the router modules by forward hooks. The extractor accepts only per-token distributions over the  $E$  experts or the router’s selected expert indices, so that other modules whose names contain “gate” are not counted as expert choices. For each model we compute the Gini coefficient of expert use pooled over layers and tokens, the same per layer (Switch: encoder layers only, since the decoder sees one token per sequence), the effective number of experts  $e^H$ , and each expert’s dominant-category share (uniform =  $1/7$ ). At the token level routing is sparse by construction: each token activates  $k$  of  $E$  experts.

### III. RESULTS

#### A. H1: Activation-Distribution Sparsity

Sparsity did not differ between groups (Table II; composite  $d = -0.26$ , 95% CI  $[-0.66, +0.13]$ ,  $p = 0.19$ ). Every index pointed, if anything, toward *less* sparse activation in the high-efficiency group, the direction opposite to the neural-efficiency prediction, and none of the seven per-task tests crossed FDR (smallest  $p_{\text{FDR}} = 0.49$ ). H1 is not supported.

#### B. H2: Task-Selective Network Engagement

Parcel-level maps show the expected anatomy (Fig. 1): WM and RELATIONAL engage lateral frontoparietal cortex, MOTOR the central sulcus and EMOTION ventral visual cortex.

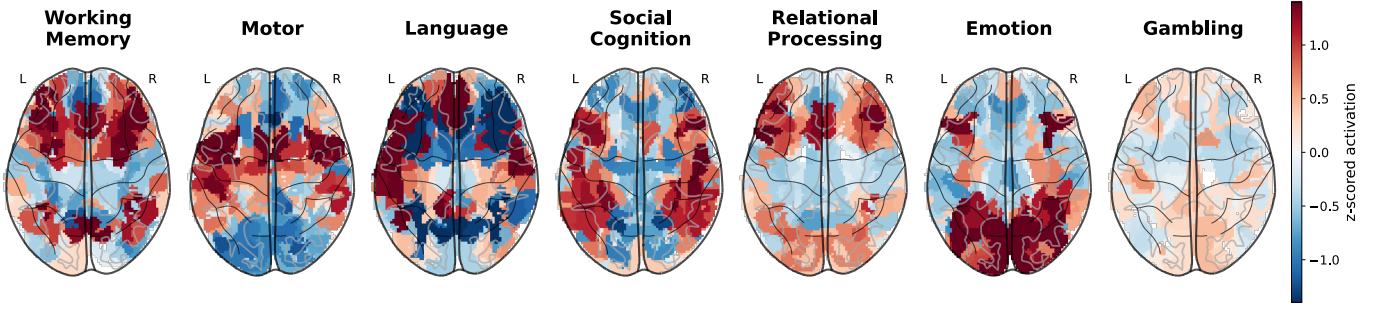


Fig. 1: Group-averaged task-evoked activation for the seven HCP tasks ( $N = 100$ ) on the HCP-MMP1.0 atlas [24] in MNI152 space (dorsal glass-brain views, one shared diverging color scale). The most engaged network matches the canonical prediction in 2/7 tasks at both Yeo-7 and 12-network resolution (WM and RELATIONAL, frontoparietal).

At the network level, mean selectivity at Yeo-7 was  $\bar{S}_t = 0.106$ , close to the uniform baseline, and the most engaged network matched the canonical prediction in 2/7 tasks (WM and RELATIONAL, both FPN). At the native 12 networks the count was also 2/7 (the same two tasks, Frontoparietal), against a chance expectation of 1.4 (permutation  $p = 0.44$ ); MOTOR peaked in Cingulo-Opercular with Somatomotor second, and LANGUAGE in Default with Language second. The signed network mean spreads focal peaks over large networks, so the match count is a coarse test, but it is not rescued by the finer partition. Selectivity did not differ between efficiency groups (Yeo-7:  $d = +0.14$ ,  $p = 0.50$ ; 12 networks:  $d = +0.40$ ,  $p = 0.050$ ). H2 is not supported.

### C. H3: Inter-Subject Routing Consistency

*Primary.* Subject-level LOO correlations were  $r = 0.680$  (high) and  $0.626$  (low):  $d = +0.52$ , 95% CI  $[+0.12, +0.92]$ , Welch  $t = 2.59$ ,  $p = 0.011$ ; Mann-Whitney  $p = 0.013$ ; bootstrap CI on the difference  $[+0.014, +0.092]$ . The label-permutation test, which needs no independence assumption, agrees ( $\Delta r = +0.069$ , null SD  $0.029$ ,  $p = 0.018$ ). H3 survives the primary BH-FDR family ( $p_{\text{FDR}} = 0.033$ ) and the four-test sensitivity family ( $p_{\text{FDR}} = 0.044$ ), and the Bayes factor gives moderate evidence for a difference ( $\text{BF}_{10} = 4.0$ ). Consistency is lower in participants who move more ( $r = -0.28$  with mean relative RMS), but motion does not differ between groups, and with motion as a covariate the group effect is unchanged ( $p = 0.013$ ). The CI is wide: effects from  $d \approx 0.1$  to  $0.9$  are compatible with the data, and resolving the lower end would need about 1100 participants per group, whereas replicating  $d = 0.52$  at 80% power needs about 60.

*The pairwise-correlation caveat.* Pooling the 1225 within-group pairwise correlations per group into an independent-samples  $t$ -test gives, on the same data,  $t = 11.68$  and  $p = 1.0 \times 10^{-30}$  with an apparent  $d = +0.47$ . The test inflates the effective sample size, from 100 participants to about 2450 pairs, and overstates significance by about 28 orders of magnitude. Fig. 2 shows why: the null distribution implied by the naive test is about  $4.9\times$  narrower than the permutation null. Here the conclusion survives correct inference, but only just; on a misspecified first-level model (event

regressors overlapping the task blocks) the same naive test gave  $p = 10^{-17}$  while subject-level inference gave  $p = 0.049$  and permutation  $p = 0.055$ , so the naive test would report an overwhelming effect in either case, regardless of whether a correct test finds one.

*Dependence on the Working Memory task.* The efficiency score is computed from Working Memory performance, and Working Memory is also one of the seven tasks in the routing profile. Removing it and recomputing H3 on the remaining six tasks reduces the group effect to  $d = +0.19$  ( $p = 0.34$ ; permutation  $p = 0.38$ ), whereas removing any one of the other six tasks leaves it in place ( $d$  from  $+0.44$  to  $+0.66$ , all Welch  $p \leq 0.032$  and permutation  $p \leq 0.046$ ). The Working Memory row alone separates the groups ( $d = +0.74$ ,  $p = 0.0004$ ). The continuous association behaves the same way:  $r = 0.36$  with all seven tasks and  $r = 0.15$  (95% CI  $[-0.05, 0.34]$ ,  $p = 0.13$ ) without Working Memory. The consistency effect is therefore carried by the task that also defines efficiency: participants who perform the 2-back task well show working-memory activation closer to the group pattern, and the other six tasks add no detectable group difference.

### D. H4: Network Modularity

Louvain found 3–6 communities per participant (median 4). Modularity did not differ between groups ( $Q$ :  $d = +0.19$ ,  $p = 0.34$ ; Fig. 3), nor did segregation ( $d = +0.35$ ,  $p = 0.085$ ) or participation ( $d = -0.14$ ,  $p = 0.50$ ; all within-family  $p_{\text{FDR}} \geq 0.25$ ). Individual partitions varied across 20 Louvain seeds (median ARI  $0.54$ , NMI  $0.54$ ), but  $Q$  was stable (within-participant SD  $0.002$  against a between-participant SD of  $0.007$ ), and seed-averaged tests ( $Q$ :  $d = +0.25$ ,  $p = 0.21$ ) and a positive-edge-only graph ( $d = +0.33$ ,  $p = 0.11$ ) reproduce the null. H4 is not supported.

### E. Sensitivity Analyses

With the continuous efficiency score in place of the median split (Table III), routing consistency, computed against the whole-sample LOO template, correlated positively with efficiency ( $r = 0.36$ , 95% CI  $[0.18, 0.52]$ ,  $p = 0.0002$ ,  $p_{\text{FDR}} = 0.001$ ; permutation of  $E_i$ :  $p = 0.0002$ ; partial

TABLE III: Summary of the brain-side hypotheses.  $d$ : high minus low efficiency, with 95% CI.  $p_{\text{FDR}}$ : primary family  $\{H1, H3, H4\}$ ;  $H2$  is in the four-test sensitivity family. TOST:  $p$  for equivalence to  $|d| < 0.5$ .  $r_E$ : Pearson correlation with the continuous efficiency score ( $p_{\text{FDR}}$  across six measures; for  $H3$  against the whole-sample LOO template).  $H3$  uses the subject-level LOO statistic; the naive pairwise test on the same data gives  $p = 1.0 \times 10^{-30}$ .  $H3$  does not survive removing the Working Memory task (Sec. III-C).

|    | Primary outcome                      | Verdict          | $d$ [95% CI]         | $p$   | $p_{\text{FDR}}$ | TOST $p$ | $\text{BF}_{10}$ | $r_E$ ( $p_{\text{FDR}}$ ) |
|----|--------------------------------------|------------------|----------------------|-------|------------------|----------|------------------|----------------------------|
| H1 | Composite sparsity (subject mean)    | Not supported    | -0.26 [-0.66, +0.13] | 0.19  | 0.29             | 0.12     | 0.45             | -0.26 (0.026)              |
| H2 | Task selectivity $\bar{S}_t$ (Yeo-7) | Not supported    | +0.14 [-0.26, +0.53] | 0.50  | 0.50             | 0.036    | 0.26             | -0.02 (0.82)               |
| H3 | LOO routing consistency              | <b>Supported</b> | +0.52 [+0.12, +0.92] | 0.011 | <b>0.033</b>     | 0.54     | 4.0              | +0.36 (0.001)              |
| H4 | Modularity $Q$                       | Not supported    | +0.19 [-0.20, +0.59] | 0.34  | 0.34             | 0.064    | 0.32             | +0.18 (0.10)               |

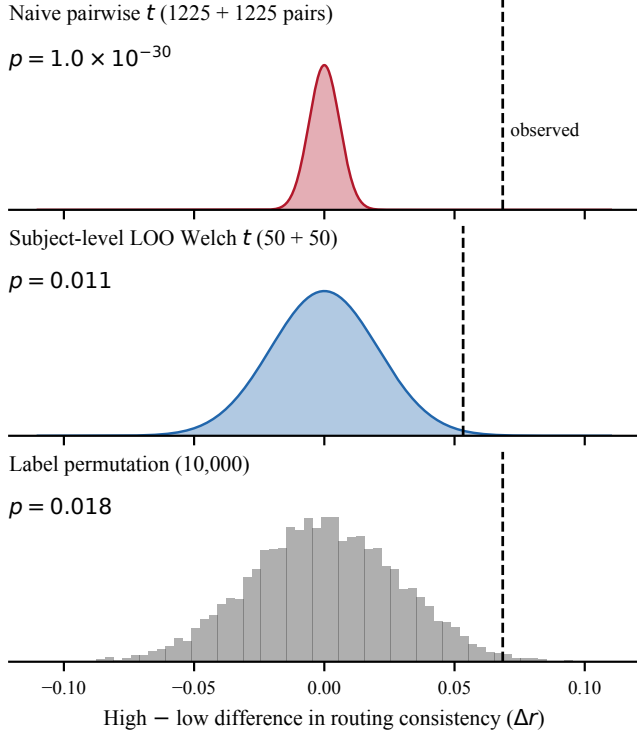


Fig. 2: Null distributions for the  $H3$  group difference on one axis. Top: normal null implied by the naive pairwise  $t$ -test. Middle: null of the subject-level LOO Welch test. Bottom: label-permutation null (10000 permutations). Dashed line: observed difference. The naive null is far too narrow because the 2450 pairs share 100 participants.

$r = 0.34$  controlling for motion). Two secondary associations also crossed FDR within this family: composite sparsity ( $r = -0.26$ ,  $p_{\text{FDR}} = 0.026$ ) and participation ( $r = -0.24$ ,  $p_{\text{FDR}} = 0.037$ ), both negative, i.e. more efficient participants distribute activation slightly more broadly and have slightly less cross-community integration; neither was detected by the pre-specified group tests. Modularity ( $r = 0.18$ ) and selectivity ( $r = -0.02$ ) did not.

#### F. MoE Routing and the Cross-Domain Comparison

Pooled over layers and tokens, all three models use their experts almost uniformly (Table IV): aggregate Gini 0.122,

TABLE IV: MoE routing. Pooled: Gini of expert use pooled over layers and tokens; per layer: mean [min, max] over layers;  $E_{\text{eff}} = e^H$  of pooled use; Dom.: mean dominant-category share (uniform  $1/7 \approx 0.143$ ).

| Model                | $k/E$ | Pooled | Per layer         | $E_{\text{eff}}$ | Dom.  |
|----------------------|-------|--------|-------------------|------------------|-------|
| Switch-8             | 1/8   | 0.122  | 0.29 [0.16, 0.45] | 7.8/8            | 0.248 |
| Qwen-MoE             | 4/60  | 0.038  | 0.17 [0.12, 0.24] | 59.9/60          | 0.247 |
| DeepSeek-16B         | 6/64  | 0.247  | 0.76 [0.33, 0.86] | 57.9/64          | 0.244 |
| Cortex (360 parcels) | —     | 0.445  | —                 | —                | —     |

0.038 and 0.247 (mean 0.135), against a cortical Gini of 0.445 over 360 parcels, a ratio of about  $3.3\times$ . The difference is not an artifact of the number of units: cortical Gini on  $k = 8, 60$  or 64 randomly chosen parcels, matching each model’s expert count, is 0.39, 0.44 and 0.44 (Fig. 4). No expert specialized for a category; the dominant-category share stayed near the uniform  $1/7$  (0.24–0.25 in all three models).

Pooling, however, hides concentration. Within a single MoE layer expert use is much more concentrated than pooled use (mean per-layer Gini 0.29, 0.17 and 0.76), because different layers favor different experts and the preferences cancel when layers are pooled; DeepSeek’s single layers are more concentrated than cortex (Fig. 4). At the token level routing is sparse by design, with  $k/E$  between 6.7% and 12.5% of experts active per token. Which system is “sparser” therefore depends on the level of comparison, and the  $3.3\times$  figure applies only to pooled expert use. We report no inferential test for the cortex–MoE difference: with only three models, a two-sample comparison of 700 subject  $\times$  task brain values against three model values has no meaningful sampling unit.

Because expert indices are arbitrary, an element-by-element correlation between the brain task-by-network matrix and a category-by-expert matrix is not interpretable. We therefore use a permutation-invariant comparison of the  $7 \times 7$  task-similarity matrix of cortex with each model’s  $7 \times 7$  category-similarity matrix (Mantel test [37] over all 5040 label orderings): Spearman  $r = -0.18, -0.41$  and  $-0.28$  (exact  $p \geq 0.77$ ). Cortex and MoE routing share no detectable category structure.

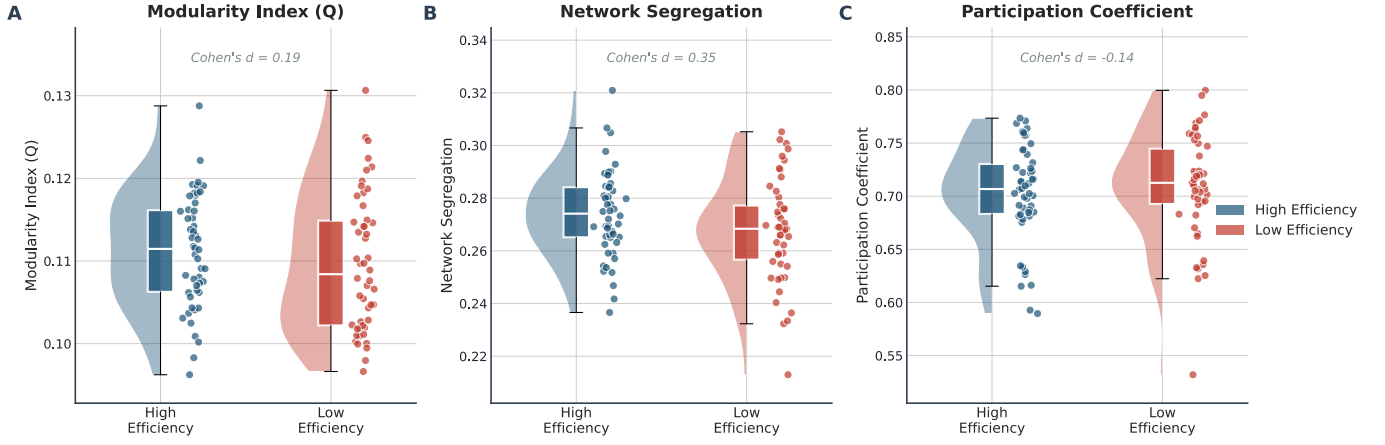


Fig. 3: Network-organization metrics by efficiency group (Louvain, seed 0): modularity  $Q$ , segregation and mean participation coefficient (half-violin, box plot and one dot per participant). Cohen’s  $d$  is printed in each panel; none of the three metrics crosses FDR.

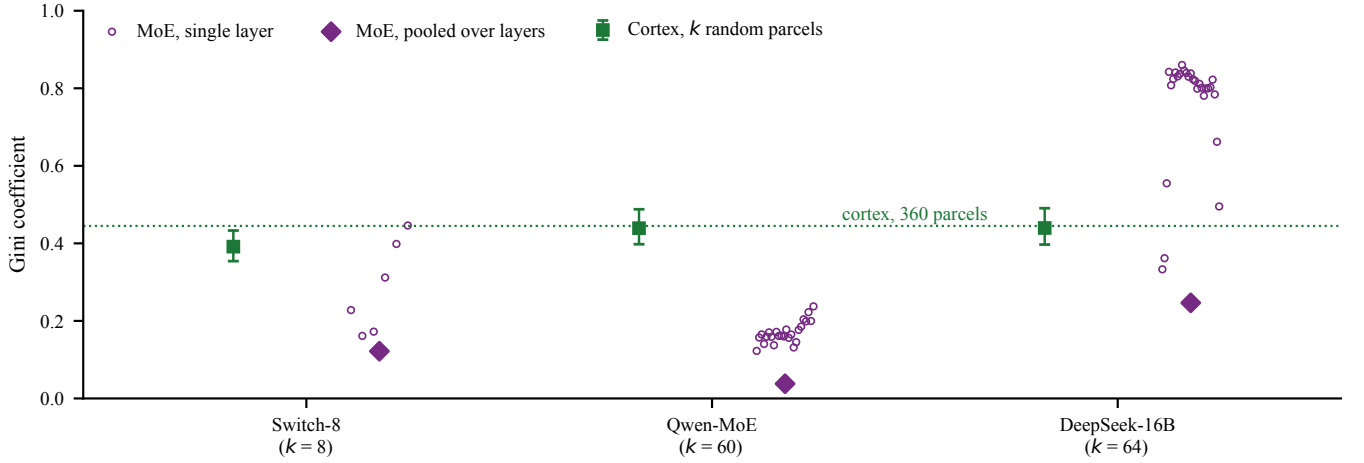


Fig. 4: Cortex versus MoE concentration at matched unit counts. Squares: cortical Gini on  $k$  randomly chosen parcels (mean and 95% range over participants  $\times$  tasks), with  $k$  equal to each model’s number of experts; dotted line: all 360 parcels. Diamonds: MoE expert use pooled over layers; open circles: single MoE layers.

#### IV. DISCUSSION

##### A. What Distinguishes Efficient Routing

Of four candidate signatures, only one survives joint FDR control: high-efficiency participants route task information across networks more like one another. The effect is moderate, is carried by both the group test and the continuous score, and holds under rank and permutation inference, which is what distinguishes it from the pairwise-correlation evidence often offered for such claims. Sparsity does not separate the groups, and the continuous analysis points away from the classical account: more efficient participants distribute activation slightly more broadly, in line with the reading of neural efficiency as targeted rather than minimal activation [2], [10]. Canonical task-network engagement is at chance even at 12 networks, and modularity is null across seeds and edge treatments, consistent with reports that modularity–cognition

associations depend on partitioning and edge weighting [13], [14]. In these data efficiency shows up in consistency, not concentration, but the consistency is task-specific. A plausible reading is that good 2-back performers engage the canonical working-memory circuitry more reliably, so their WM maps converge on the group pattern; this links performance to the typicality of activation within one task and is not evidence that efficient individuals route information more consistently across the task battery.

##### B. Implications for Routing-Level Biomarkers

Routing consistency is a candidate marker, not a validated one. Its CI spans small to large effects, it comes from one cohort, and it disappears when the task that defines efficiency is removed. A replication should define efficiency outside the imaged battery, or hold the defining task out of the routing

profile; with about 60 participants per group it would test the observed effect, while ruling out a small one needs far more. Any such replication, and any claim of “more consistent routing” in preclinical Alzheimer’s, schizophrenia [3], [4], [6] or rehabilitation cohorts, should report a subject-level or permutation test: on our data the naive pairwise test overstates significance by about 28 orders of magnitude, and on a misspecified first-level model it reported  $p = 10^{-17}$  for an effect that subject-level inference placed at  $p \approx 0.05$ .

#### C. What the MoE Comparison Does and Does Not Show

Pooled over layers, MoE expert use is near-uniform and category-blind, as expected from load-balancing objectives [16], but single layers can be more concentrated than cortex. Moreover, our efficiency signal lies in the consistency of routing across individuals, which has no direct analogue in a single trained model; the cross-domain comparison therefore concerns concentration, not the property that distinguishes efficient brains here.

#### D. Limitations

(i) Family structure is not modelled; HCP-YA siblings and twins may make participants non-independent, which could inflate the H3 evidence. Consistency also depends on head motion, which we controlled for statistically but not by design. (ii) The data were not ICA-FIX denoised, and no motion threshold was applied, although motion was low and balanced across groups. (iii) The efficiency score comes from one session and one of the seven tasks, and that task carries the H3 effect (Sec. III-C); its run-to-run reliability is 0.80. (iv) H4 uses task-profile similarity with seven observations per edge, not time-series FC, and results may differ with Leiden refinement [38]. (v) The task–category mapping (Table I) is approximate, and the MoE results apply to the three examined architectures. (vi) The sample is a narrow healthy young-adult cohort. The pairwise-correlation pitfall is a property of the statistical procedure and does not depend on these.

#### V. CONCLUSION

In 100 HCP young adults, joint FDR-controlled testing of four routing-level signatures of cognitive efficiency supports one: high-efficiency participants route task information more consistently with one another ( $d = +0.52$ ,  $p_{\text{FDR}} = 0.033$ ), a result that holds under permutation, rank, continuous-score and Bayesian analysis but is carried by the Working Memory task that also defines efficiency (without it,  $d = +0.19$ ,  $p = 0.34$ ). Sparsity, canonical task-network engagement and modularity are not supported. The conventional pairwise-correlation  $t$ -test would have reported the consistency effect at  $p = 10^{-30}$ ; subject-level and permutation inference put it at  $p \approx 0.01$ – $0.02$ , and on a flawed first-level model the naive test was just as confident while correct inference was not. Cortex is more concentrated than pooled MoE expert use, but not than every single MoE layer. Routing consistency is at most a candidate marker of efficient cognition; testing it requires an efficiency measure that is independent of the imaged tasks, and subject-level inference.

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#### CODE AND DATA AVAILABILITY

Analysis code, group-level results and the MoE routing outputs are available at <https://github.com/xiaoyanLi629/routing-signatures-efficiency>. HCP data are available from the Human Connectome Project under its open-access terms and are not redistributed.

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